SIDDHARTH SHAH

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Data journalist and newsroom technologist - seeking a work opportunity in India starting May 2026, completing a U.S. B.S. at Penn State with 130+ credits across Computer Engineering, Applied Data Sciences, and IST. Built production ML pipelines, cloud ETL systems, and real-time analytics dashboards across 3 professional roles in the U.S., grew up on Zakaria, Mark Tully, Prannoy Roy, and Sudhir Chaudhary long before writing a line of code. Seeking a Summer 2026 internship (12+ weeks) followed by a transition to full-time work at India Global Review to build the data infrastructure, analytical storytelling, and audience intelligence a digital-first global newsroom needs from day one.

EDUCATION

Pennsylvania State University · *State College, PA, USA* · GPA : 3.7/4.0 — Aug 2021 – Aug 2026

B.S. Information Sciences & Technology (Prior: Computer Engineering; Applied Data Sciences — 130+ credits in-residence at University Park)

Relevant: Machine Learning · Statistical Modeling · Database Systems · Human-Centered Design · Software Engineering · Linear Algebra · Philosophy (writings recognized by faculty)

PROFESSIONAL EXPERIENCE

University Research Associate · *ReSense Lab — Penn State University* — Oct 2024 – Aug 2025 · University Park, PA

- Reduced false-positive rate by **15%** by engineering real-time geospatial algorithms in Python, the same pipeline design that powers map-based election and conflict-zone visualisations in modern newsrooms.
- Mined **15,000+** telemetry records with Pandas and NumPy clustering to identify 8 behavioural event categories, a dataset scale and analytical method directly transferable to audience-behaviour and engagement analytics.
- Accelerated system response time by **20%** by rebuilding real-time data visualisation pipelines (Matplotlib, custom tooling), the low-latency dashboard architecture newsrooms need for live election and breaking-news coverage.

Data Analyst Intern · *Matriosh* — Jun 2024 – Aug 2024 · New York City, NY

- Eliminated **40%** of manual data-processing effort by engineering automated ETL pipelines in Azure Databricks and Splunk across 5+ enterprise data streams, the exact infrastructure pattern for ingesting and cleaning newsroom data at scale.
- Delivered **4 executive-facing** Power BI and Tableau dashboards tracking KPIs and anomaly detection, directly applicable to editorial performance dashboards, audience metrics, and content reach analytics.
- Generated strategic intelligence via Python web scraping and SQL across **3 data sources**, producing actionable insights that shaped quarterly strategy, mirroring the research workflow behind data-driven investigative journalism.

Applications Developer & Business Analyst Intern · *AllianceTek Inc.* — Jun 2023 – Aug 2023 · Malvern, PA

- Cut API latency and page-load time by **30%** by re-architecting Flask APIs and React/TypeScript frontend, the full-stack velocity a digital-first media platform needs to ship features at startup pace.
- Translated stakeholder requirements into UX wireframes across 3 sprint cycles, reducing design rework by **25%**, the cross-functional communication skill that bridges editorial and engineering teams.
- Clustered feature-usage patterns with Pandas and PostgreSQL across **2,000+** interaction records to guide product roadmap, the same analytical lens applicable to content performance and audience segmentation.

DATA JOURNALISM & EDITORIAL WORK

Fiscal Federalism in India — Original Data Investigation — Apr 2026

Python · Pandas · Plotly · RBI & Finance Commission datasets · 800-word analytical essay

- Quantified the cross-subsidisation cycle between India's northern and southern states using RBI State Finance data, producing a **4-chart data story** with an opinionated 800-word essay, published at siddharthgshah.com/work.

Complalg — AI-Native Compliance Platform — Dec 2025 – Present

NestJS · FastAPI · LLM/RAG · pgvector · Knowledge Graph · Chrome Extension

- Architected a knowledge-graph + LLM-RAG backend for automated regulatory filing and trade surveillance; Chrome Extension cut form-submission time by **40%** and surfaced 25% more high-risk compliance events in QA.

Forensic Image Analysis (CNN / Deep Learning) — Sep – Dec 2022

TensorFlow · Keras · CNN · Large-Scale Image Datasets

- Led 10-engineer team to build a tamper-detection and object-recognition pipeline achieving **77%** accuracy, reducing investigation triage time by 35%, a workflow directly relevant to media verification and fact-checking.

BEYOND THE ENGINEERING

Writing: Philosophical essay on autonomy and the human condition ("Into the Wild" / McCandless) recognised by Penn State faculty and displayed in department. **Photography:** Avid photographer with an eye for cityscapes and night skies, portfolio at siddharthgshah.com. **Current Affairs:** Lifelong follower of geopolitics and international economics; can identify most heads of state from memory; news consumption since grade school (Zakaria, Amanpour, Tully, Roy, Dua, Palki Sharma). **Languages:** English (Fluent) · Hindi (Native) · Gujarati (Native) · French (Conversational) · Sanskrit.

TECHNICAL SKILLS

Data & Analytics: Python (Pandas, NumPy, Scikit-learn) · Power BI · Tableau · Excel · ETL pipelines · Azure Databricks · Splunk · PostgreSQL · MySQL · Web scraping

AI / ML: TensorFlow · Keras · Scikit-learn · NLP (NLTK) · LLM/RAG agents · Deep learning · Clustering & classification · Model evaluation

Engineering: Python · TypeScript / Node.js · Java · C++ · Go · SQL · Flask · FastAPI · NestJS · React · Docker · CI/CD · Git